

BACKEND SCHEMA

Mehmoodah Academy — School Operating System (V1)

Version: 1.0

Backend: Supabase

Database: PostgreSQL

Auth: Supabase Auth

Purpose: Define backend architecture, tables, APIs, auth rules, storage, and access control.

1. BACKEND ARCHITECTURE

CLIENT (Flutter Web)

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SUPABASE AUTH

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API / EDGE FUNCTIONS

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POSTGRES DATABASE

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STORAGE

↓

REALTIME

Architecture Type:

BaaS + Modular Backend

2. AUTHENTICATION

Provider: Email + Password

Flow:

Login

↓

Auth Verify

↓

Get Role

↓

Open Dashboard

Session:

7 Days

Refresh:

Automatic

Roles:

ADMIN

TEACHER

STUDENT

3. AUTH USERS

Managed by:

auth.users

Fields:

id

email

created_at

Internal Only

Never edit directly.

4. PUBLIC.USERS

Purpose: Application Users

TABLE: users

Columns:

id UUID PK

auth_user_id UUID

email TEXT

role TEXT

active BOOLEAN

created_at

updated_at

Rules:

email UNIQUE

role required

Values:

admin

teacher

student

5. TEACHERS

TABLE: teachers

Columns:

id UUID PK

user_id FK

teacher_code

full_name

phone

subject

joining_date

status

Indexes:

teacher_code

user_id

6. STUDENTS

TABLE: students

Columns:

id UUID PK

user_id FK

student_code

full_name

class_id FK

roll_number

guardian_name

phone

dob

status

Indexes:

student_code

roll_number

7. CLASSES

TABLE: classes

Columns:

id UUID PK

name

section

teacher_id FK

academic_year

status

Example:

Grade 8

8. ATTENDANCE

TABLE: attendance

Columns:

id UUID PK

student_id FK

class_id FK

teacher_id FK

date

status

created_at

Allowed:

present

absent

late

leave

Constraints:

UNIQUE (student_id,date)

9. EXAMS

TABLE: exams

Columns:

id UUID PK

class_id FK

title

term

published

created_at

10. RESULTS

TABLE: results

Columns:

id UUID PK

exam_id FK

student_id FK

marks

percentage

grade

remarks

Constraints:

UNIQUE

exam_id

student_id

11. ASSIGNMENTS

TABLE: assignments

Columns:

id UUID PK

teacher_id FK

class_id FK

title

description

attachment

due_date

status

12. SUBMISSIONS

TABLE: submissions

Columns:

id UUID PK

assignment_id FK

student_id FK

submission_url

submitted_at

status

13. TIMETABLE

TABLE: timetable

Columns:

id UUID PK

class_id FK

teacher_id FK

subject

day

start_time

end_time

Validation:

No overlap

14. ANNOUNCEMENTS

TABLE: announcements

Columns:

id UUID PK

title

content

target_role

published

created_at

Values:

all

teachers

students

15. STORAGE

BUCKETS

student-photos

teacher-photos

assignments

submissions

reports

Structure:

bucket/

year/

month/

file

Example:

assignments/

2026/

assignment.pdf

16. ROW LEVEL SECURITY

users

Admin: ALL

Teacher: SELF

Student: SELF

teachers

Admin: CRUD

Teacher: READ OWN

Student: NONE

students

Admin: CRUD

Teacher: READ ASSIGNED

Student: READ OWN

attendance

Admin: ALL

Teacher: ASSIGNED

Student: READ OWN

results

Admin: ALL

Teacher: CLASS

Student: READ OWN

assignments

Admin: ALL

Teacher: OWN

Student: READ

17. API ENDPOINTS

AUTH

POST

/auth/login

POST

/auth/logout

POST

/auth/reset

USERS

GET

/users

POST

/users

TEACHERS

GET

/teachers

POST

/teachers

PUT

/teachers/:id

DELETE

/teachers/:id

STUDENTS

GET

/students

POST

/students

PUT

/students/:id

DELETE

/students/:id

ATTENDANCE

GET

/attendance

POST

/attendance

RESULTS

GET

/results

POST

/results

ASSIGNMENTS

GET

/assignments

POST

/assignments

REPORTS

GET

/reports

18. EDGE FUNCTIONS

create_teacher()

create_student()

publish_results()

generate_report()

upload_assignment()

19. REALTIME EVENTS

attendance.created

results.updated

assignment.created

announcement.sent

20. BACKUP

Database: Daily

Storage: Weekly

Retention: 30 Days

21. ENVIRONMENT VARIABLES

SUPABASE_URL

SUPABASE_ANON_KEY

SUPABASE_SERVICE_KEY

FIREBASE_KEY

APP_ENV

22. DEPLOYMENT

GitHub

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Vercel

↓

Supabase



Production

END OF BACKEND SCHEMA